

Wi-Fi Roaming Experiment – Packet Capture Analysis

1. Introduction

Wi-Fi roaming is the process in which a wireless client moves from one Access Point to another when the signal from the currently connected AP becomes weak.

The purpose of this experiment was to understand how a client behaves when the wireless conditions of an Access Point are changed and to observe this behaviour using raw 802.11 packet captures.

An Airtel Wi-Fi router was used as the Access Point, an iPhone was used as the client, and a personal laptop running Ubuntu Live was used to capture wireless packets in monitor mode.

Two captures were taken:

- **Capture 1 – Normal/Auto configuration**
- **Capture 2 – Reduced TX Power and Bandwidth**

The captures were then analyzed in Wireshark.

2. Concept Behind the Experiment

When a Wi-Fi client is connected to an AP, it continuously communicates with that AP.

If the client moves farther away, the wireless signal becomes weaker. In a Wi-Fi environment with multiple APs, the client can look for another suitable AP and move its connection to that AP.

The basic idea is:

Connected AP → Signal becomes weak → Client discovers another AP → Client changes AP → Communication continues

The important information for identifying roaming in a packet capture is the **BSSID**.

If the client changes from:

BSSID 1 → BSSID 2

then a possible roaming event can be investigated.

Reassociation frames are also important because they are specifically associated with a client moving its association to another AP.

3. First Capture – Normal/Auto Configuration

Initially, the Airtel router was kept in its normal/default configuration.

The iPhone was connected to the Airtel Wi-Fi and the packet capture was started.

The first capture was saved as:

my_capture.cap

The Airtel AP was identified by:

BSSID: FC:DD:55:B8:DA:E2

The capture shows the normal connection and communication between the client and the AP.

WLAN Advanced Settings

Working band:	2.4GHz
Channel:	Auto
Current Channel:	11
AP isolation:	Off
Wi-Fi bandwidth:	Auto
Sleep time:	10 Minutes
Tx Power:	Auto
Max User:	10

4. Changing the Wireless Conditions

After completing the first capture, the Airtel router configuration was changed.

The **TX Power was reduced**, and the **channel bandwidth was also reduced**.

Reducing transmit power reduces the effective coverage of the AP.

Reducing bandwidth changes the channel-width capability of the AP.

The second capture was then taken using the changed configuration.

The second capture was saved as:

afterchange-01.cap

WLAN Advanced Settings

Working band:	2.4GHz
Channel:	11
Current Channel:	11
AP isolation:	Off
Wi-Fi bandwidth:	20M
Sleep time:	10 Minutes
Tx Power:	Low
Max User:	10

5. Effect Visible in the Beacon

One useful observation can be made directly from the Airtel Beacon.

In the first capture, the AP advertises support for **40 MHz channel width**.

After changing the bandwidth, the corresponding capability is no longer advertised in the second capture.

The primary channel remains **Channel 11**.

This shows that the configuration change made on the router is reflected in the information advertised by the AP.

```
▶ Tag: Extended Supported Rates 24, 36, 48, 54, [Mbit/sec]
▼ Tag: HT Capabilities
  Tag Number: HT Capabilities (45)
  Tag length: 26
  ▼ HT Capabilities Info: 0x106e
    .... = HT LDPC coding capability: Transmitter does not support receiving LDPC coded packets
    ....1. = HT Support channel width: Transmitter supports 20MHz and 40MHz operation
    ....11.. = HT SM Power Save: SM Power Save disabled (0x3)
    ....0.... = HT Green Field: Transmitter is not able to receive PPDU's with Green Field (GF) preamble
    ....1. .... = HT Short GI for 20MHz: Supported
    ....1. .... = HT Short GI for 40MHz: Supported
    ....0... .... = HT Tx STBC: Not supported
    ....00 .... = HT Rx STBC: No Rx STBC support (0x0)
    ....0... .... = Reserved: 0x0
    ....0... .... = HT Max A-MSDU length: 3839 bytes
    ....1 .... = HT DSSS/CCK mode in 40MHz: Will/Can use DSSS/CCK in 40 MHz
    ....0. .... = Reserved: 0x0
    ....0... .... = HT Forty MHz Intolerant: Use of 40 MHz transmissions unrestricted/allowed
    ....0... .... = Reserved: 0x0
```

```

  ▶ Tag: Extended Supported Rates 24, 36, 48, 54, [Mbit/sec]
  ▼ Tag: HT Capabilities
    Tag Number: HT Capabilities (45)
    Tag length: 26
    ▼ HT Capabilities Info: 0x102c
      .... 0 = HT LDPC coding capability: Transmitter does not support receiving LDPC coded ...
      .... 0 = HT Support channel width: Transmitter only supports 20MHz operation
      .... 11.. = HT SM Power Save: SM Power Save disabled (0x3)
      .... 0 = HT Green Field: Transmitter is not able to receive PPDU's with Green Field (GF...)
      .... 1. .... = HT Short GI for 20MHz: Supported
      .... 0.. .... = HT Short GI for 40MHz: Not supported
      .... 0... .... = HT Tx STBC: Not supported
      .... 00 .... = HT Rx STBC: No Rx STBC support (0x0)
      .... 0.. .... = Reserved: 0x0
      .... 0... .... = HT Max A-MSDU length: 3839 bytes
      ...1 .... .... = HT DSSS/CCK mode in 40MHz: Will/Can use DSSS/CCK in 40 MHz

```

6. Deauthentication and Reconnection

The captures contain Deauthentication frames.

After a connection is terminated, the client can establish the connection again using:

Authentication → Association → EAPOL → Data

This sequence can therefore be used to understand client reconnection behaviour.

No.	Time	Source	Destination	Protocol	Length	Info
1276	5.301058	f2:f6:06:48:fd:55	WeWinswirele_b8:da:...	802.11	26	Deauthentication, SN=478, FN=0, Flags=.....
1340	10.430372	f2:f6:06:48:fd:55	WeWinswirele_b8:da:...	802.11	154	Association Request, SN=2101, FN=0, Flags=....., SSID="Airtel-MyWiFi-AMF-311WW-DAE2"
3313	14.097191	f2:f6:06:48:fd:55	WeWinswirele_b8:da:...	802.11	26	Deauthentication, SN=2107, FN=0, Flags=.....
3372	18.857576	f2:f6:06:48:fd:55	WeWinswirele_b8:da:...	802.11	154	Association Request, SN=798, FN=0, Flags=....., SSID="Airtel-MyWiFi-AMF-311WW-DAE2"
4787	21.606733	f2:f6:06:48:fd:55	WeWinswirele_b8:da:...	802.11	26	Deauthentication, SN=804, FN=0, Flags=.....
4848	25.856343	f2:f6:06:48:fd:55	WeWinswirele_b8:da:...	802.11	154	Association Request, SN=1187, FN=0, Flags=....., SSID="Airtel-MyWiFi-AMF-311WW-DAE2"
6406	29.182947	f2:f6:06:48:fd:55	WeWinswirele_b8:da:...	802.11	26	Deauthentication, SN=1193, FN=0, Flags=.....
6566	38.017061	f2:f6:06:48:fd:55	WeWinswirele_b8:da:...	802.11	154	Association Request, SN=2388, FN=0, Flags=....., SSID="Airtel-MyWiFi-AMF-311WW-DAE2"
7645	46.696237	f2:f6:06:48:fd:55	WeWinswirele_b8:da:...	802.11	26	Deauthentication, SN=2393, FN=0, Flags=.....
7719	51.005706	f2:f6:06:48:fd:55	WeWinswirele_b8:da:...	802.11	154	Association Request, SN=3969, FN=0, Flags=....., SSID="Airtel-MyWiFi-AMF-311WW-DAE2"

No.	Time	Source	Destination	Protocol	Length	Info
1276	5.301058	f2:f6:06:48:fd:55	WeWinswirele_b8:da:...	802.11	26	Deauthentication, SN=478, FN=0, Flags=.....
3313	14.097191	f2:f6:06:48:fd:55	WeWinswirele_b8:da:...	802.11	26	Deauthentication, SN=2107, FN=0, Flags=.....
4787	21.606733	f2:f6:06:48:fd:55	WeWinswirele_b8:da:...	802.11	26	Deauthentication, SN=804, FN=0, Flags=.....
6406	29.182947	f2:f6:06:48:fd:55	WeWinswirele_b8:da:...	802.11	26	Deauthentication, SN=1193, FN=0, Flags=.....
7645	46.696237	f2:f6:06:48:fd:55	WeWinswirele_b8:da:...	802.11	26	Deauthentication, SN=2393, FN=0, Flags=.....

12. What the Experiment Shows

The experiment demonstrates how changing the AP's radio configuration can affect client connectivity.

The captures show:

AP Advertisement → Authentication → Association → WPA2 EAPOL → Data → Deauthentication → Reconnection

The second capture also shows the change in the AP's advertised channel-width capability.

The physical experiment showed that when the iPhone moved away from the Airtel router, it eventually lost or changed its Wi-Fi connection and connected to another available network.

However, because the second network was the father's independent hotspot and the capture does not show a BSSID transition or Reassociation sequence, the capture does **not provide direct evidence of conventional AP-to-AP roaming**.

Observation	Capture 1: Normal / Auto	Capture 2: Low Power + Reduced Bandwidth
Total packets captured	9,459	6,594
Beacon frame — 40 MHz width advertised (HT Capabilities IE)	YES — supported	NO — not supported
HT Operation IE — channel in use	Primary ch 11, 20 MHz active	Primary ch 11, 20 MHz active
Full WPA2 4-way EAPOL handshakes seen	5 complete handshakes (20 EAPOL frames)	2 complete handshakes (8 EAPOL frames)
Deauthentication frames seen	5	2
Probe Requests / Responses	5 / 154	2 / 65
Authentication frames	10	4
Association Req/Resp pairs	5 / 5	2 / 2
Security type (unchanged)	WPA2-PSK (RSN) + WPA1, CCMP/TKIP	WPA2-PSK (RSN) + WPA1, CCMP/TKIP

13. Conclusion

This experiment helped me understand Wi-Fi roaming and, more importantly, how roaming-related behaviour can be verified using actual 802.11 packets.

By changing the Airtel router's transmit power and bandwidth, I created different wireless conditions and captured the resulting client behaviour.

The packet analysis allowed me to identify the major stages of the Wi-Fi connection:

Authentication → Association → WPA2 EAPOL → Data

I also observed Deauthentication and subsequent reconnection sequences.

The Beacon comparison showed that the bandwidth configuration change was reflected in the AP's advertised HT capabilities.

For roaming specifically, I checked the BSSID and Reassociation frames. Since the supplied captures remain associated with the same Airtel BSSID and do not contain a Reassociation sequence to another AP, I cannot claim that an AP-to-AP roaming event was captured.

The main learning from the experiment is that **roaming must be identified from the client's change of AP/BSSID and the corresponding 802.11 management-frame sequence, rather than simply from a loss and re-establishment of connectivity**.